DevOps Infrastructure – Unit 2 Notes (Part 6)

Registries: Docker Hub, GitHub Container Registry and Private Registries

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1. Docker Registries – Overview

Definition

A registry is a storage system for Docker images.

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2. Docker Hub

Definition

Docker Hub is the default public registry used by Docker.

Features

- Public repositories
- Official images (nginx, ubuntu, mysql)
- Easy integration with Docker CLI

Example

docker pull nginx

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3. GitHub Container Registry (GHCR)

Definition

GHCR is a container registry provided by GitHub.

Features

- Stores Docker images
- Integrated with GitHub repositories
- Supports private and public images

Use Case

- CI/CD pipelines
- Version-controlled deployments

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4. Private Registries

Definition

Private registries are secure repositories used within organizations.

Examples

- AWS Elastic Container Registry (ECR)
- Google Container Registry (GCR)
- Azure Container Registry (ACR)

Benefits

- Security and access control
- Internal usage
- Custom configurations

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5. Image Push and Pull

Push Image

```
docker push my_app:1.0
```

Pull Image

```
docker pull my_app:1.0
```

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6. Summary

- Registries store and distribute images
- Docker Hub is public and widely used
- GHCR integrates with GitHub workflows
- Private registries offer security and control